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QUESTION: Write a C program to perform the following menu-driven operations on a given 2D array:

1. Find the sum of all elements in a matrix.
2. Find the sum of each row of a matrix and store it in an array.
3. Find the transpose of a matrix.
4. Exit

Sample Output:

Enter the number of rows and columns: 2 3

Enter the matrix elements:

1 2 3

4 5 6

Enter your choice: 1
Sum of all elements = 21

Enter your choice: 2

Row sums:

Sum of row 1 = 6

Sum of row 2 = 15

The row-sum array contains: [6, 15]

Enter your choice: 3

Transpose of the matrix:

1 4

2 5

3 6

SOLUTION:

1. Aim

To write a menu-driven C program using a switch statement to perform various operations on a 2D array (matrix), including finding the **sum of all elements**, calculating and storing the **sum of each row in a 1D array**, and finding the **transpose of the matrix**.

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2. Algorithm

Main Algorithm

1. Start the program.
2. Input the number of rows and columns of the matrix.
3. Input the matrix elements using nested loops.
4. Display the following menu:
    - Find the sum of all matrix elements.
    - Find the sum of each row and store it in an array.
    - Find the transpose of the matrix.
    - Exit.
5. Accept the user's choice.
6. Use a switch statement to perform the selected operation.
7. Repeat the menu until the user selects **Exit**.
8. Stop.

Operation 1: Sum of All Elements

1. Initialize sum = 0.
2. Traverse all the elements of the matrix using nested loops.
3. Add each matrix element to sum.
4. Display the final value of sum.

Operation 2: Sum of Each Row

1. Declare a 1D array rowSum to store the sum of each row.
2. Traverse the matrix row by row.
3. For each row, initialize its sum to 0.
4. Add all the elements of that row.
5. Store the calculated sum in rowSum[i].
6. Repeat for all rows.

7. Display the sum of each row.
8. Display the complete row-sum array.

Operation 3: Transpose of Matrix

1. Traverse the original matrix using nested loops.
2. Interchange the row and column indices while displaying the elements.
3. For an element matrix[i][j], the corresponding element in the transpose is matrix[j][i].
4. Therefore, a matrix of size rows × cols produces a transpose of size cols × rows.
5. Display the transposed matrix.

C PROGRAM:

```
#include <stdio.h>


#define MAX 100


int main()
{
   int matrix[MAX][MAX];
   int rowSum[MAX];
   int rows, cols;
   int i, j, choice, sum;


   printf("Enter the number of rows and columns: ");
   scanf("%d %d", &rows, &cols);


   printf("\nEnter the matrix elements:\n");


   // Input matrix elements
   for(i = 0; i < rows; i++)
```

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```
{
   for(j = 0; j < cols; j++)
   {
      scanf("%d", &matrix[i][j]);
   }
}

do
{
   printf("\n----- MATRIX OPERATIONS MENU -----\n");
   printf("1. Sum of all elements\n");
   printf("2. Sum of each row\n");
   printf("3. Transpose of matrix\n");
   printf("4. Exit\n");

   printf("\nEnter your choice: ");
   scanf("%d", &choice);

   switch(choice)
   {
      case 1:
      {
         sum = 0;

         // Calculate sum of all elements
         for(i = 0; i < rows; i++)
         {
```

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```
      for(j = 0; j < cols; j++)
      {
         sum = sum + matrix[i][j];
      }
   }

   printf("Sum of all elements = %d\n", sum);

   break;
}

case 2:
{
   // Calculate sum of each row
   for(i = 0; i < rows; i++)
   {
      rowSum[i] = 0;

      for(j = 0; j < cols; j++)
      {
         rowSum[i] = rowSum[i] + matrix[i][j];
      }
   }

   printf("\nRow sums:\n");

   for(i = 0; i < rows; i++)
```

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```
    {
        printf("Sum of row %d = %d\n",
                i + 1, rowSum[i]);
    }

    // Display row-sum array
    printf("\nThe row-sum array contains: [");

    for(i = 0; i < rows; i++)
    {
        printf("%d", rowSum[i]);

        if(i < rows - 1)
        {
            printf(", ");
        }
    }

    printf("]\n");

    break;
}

case 3:
{
    printf("\nTranspose of the matrix:\n");
```

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```
            // Display transpose
            for(j = 0; j < cols; j++)
            {
               for(i = 0; i < rows; i++)
               {
                  printf("%d\t", matrix[i][j]);
               }

               printf("\n");
            }

            break;
         }

         case 4:
            printf("Program exited successfully.\n");
            break;

         default:
            printf("Invalid choice! Please try again.\n");
      }

   } while(choice != 4);

   return 0;
}
```

TEST CASE TABLE:

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Test Case	Matrix	Operation	Expected Result
1	1 2 3 / 4 5 6	Sum of all elements	21
2	1 2 3 / 4 5 6	Row sums	[6, 15]
3	1 2 3 / 4 5 6	Transpose	1 4 / 2 5 / 3 6
4	5	Sum of all elements	5
5	5	Row sum	[5]
6	1 2 / 3 4	Transpose	1 3 / 2 4
7	Any valid matrix	Exit	Program terminates
8	Any valid matrix	Invalid menu choice	Displays invalid choice message

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COMPILATION AND EXECUTION:



